PROBLEM STATEMENT #41 | Developer Tools & IT Operations

OpenAPI Mock Server Generator

1. Problem Context & Overview

A developer productivity tool that ingests OpenAPI 3.0 YAML/JSON specifications, creates dynamic mock REST endpoints with schema validation, and simulates configurable response latency.

Target Stakeholders / Actors: *API Developer, QA Engineer*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall parse OpenAPI 3.0 schema definitions and generate dynamic HTTP mock endpoints returning randomized schema-compliant JSON payloads.
Sample Acceptance Criteria: Pass: Mock endpoint responds with correct JSON schema types; Fail: Invalid endpoint path returns 200 OK.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: The mock server engine must sustain 1,000 mock API requests per second with user-configured simulated latency accurate within ± 10 ms.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.